Invariance is not robustness: what predicts adversarial robustness, from CIFAR classifiers to CLIP encoders

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Abstract

A model that assigns the same label to a shifted view of an image is often called robust, and this shift-invariance is offered as an explanation for, or a route to, adversarial robustness. We show that this is the wrong lens, and identify what predicts robustness instead: a margin-to-sensitivity ratio measured in the attack's own norm, and the shape of the gradient behind it. Invariance is about ignoring changes that should not matter; adversarial robustness is about not being flipped by a small worst-case change. These are different directions, and the invariance metric the field tracks, shift-consistency, is *saturated* on modern models: across sixteen frozen vision-language encoders it ranges only from 0.96 to 0.99, robust or not, so it has no room left to order them. A threat-matched margin-to-Lipschitz ratio η/L, the classification margin divided by the dual-norm size of its input gradient, does order robustness, from small CIFAR classifiers to a ResNet and, per image, within frozen CLIP encoders, and it is a certificate we complete to a two-sided bracket. The much-cited "invariance helps robustness" effect is a weak-attack artifact: it appears under the fast gradient sign method and vanishes under AutoAttack. We then close the one gap this diagnostic leaves. The ratio *detects* robustness but does not *rank* it among already-robust models; the missing ingredient is the *shape* of the gradient, its anisotropy $A = \|\nabla M\|_1/\|\nabla M\|_2$, which ranks thirty official RobustBench models at Spearman -0.79, and -0.77 after controlling for clean accuracy. We identify the mechanism, an effective-dimension margin deficit, and separate it from the curvature account. Finally, the saturation is modality-general: on zero-shot CLIP both visual (shift) and textual (paraphrase) invariance saturate, and η/L predicts the image-adversarial axis while neither consistency metric does. Throughout we use AutoAttack as the arbiter and check for gradient masking.

1 Introduction

Call an image classifier *shift-invariant* if its prediction is unchanged by translations of the input. Two literatures disagree about what this buys. Ge et al. [4] show, for a binary problem in which each class is one image and all its shifts, that a fully shift-invariant network is fooled by an ℓ_2 perturbation of order $1/\sqrt{d}$ while a fully connected network needs order 1: invariance *reduces* robustness. A separate architectural line points the other way. Anti-aliased pooling raises shift-consistency, how often a shift leaves the prediction unchanged, along with clean accuracy and corruption robustness [17], and reducing aliasing has since been reported to raise *adversarial* robustness too [5, 11]: invariance *helps*. A third line proves a trade-off between geometric and adversarial robustness [7, 13].

These statements are not contradictory. They are regimes of one quantity, a margin-to-Lipschitz ratio. For an input x with label y, let the margin be $M(x) = f_y(x) - \max_{j\neq y} f_j(x)$, the gap between the correct logit and the strongest competitor. The per-image ratio is $M(x)/\|\nabla_x M(x)\|_q$, the margin divided by how fast it changes when the input is perturbed, measured in the norm dual to the attack

(q=1 for an ℓ_∞ attack, q=2 for ℓ_2). A first-order argument gives a certificate: the adversarial radius of x is at least this ratio, because a perturbation of size δ moves the margin by at most $\|\nabla_x M\|_q \|\delta\|$. For a whole model we summarize with η/L, the mean margin η over the mean gradient size L on correctly classified clean images. We always match the norm to the attack and call this the *threat-matched* ratio. It has two properties that make it the right scalar. It needs no attack to compute. And it is *gauge-free*: rescaling the logits $f \mapsto cf$ multiplies η and L by the same c, so the ratio, and the input-space radius it lower-bounds, are unchanged. A model's logit temperature cannot inflate or deflate it.

The reason invariance and robustness come apart is geometric. Invariance is being flat *along* the shift orbit, a direction that stays inside the correct class region. Robustness is the *distance* to the decision boundary, a different, near-orthogonal direction. A model can have a long flat orbit (very invariant) and still sit a hair from the boundary (fragile). We call this the *dissociation*, and it is the paper's organizing observation: shift-invariance stays high and flat while adversarial robustness varies widely across models, so one cannot be read off the other. Formally, imposing invariance projects the classes onto the invariant subspace, and robustness is a distance to the boundary (Figure 1); measuring one says little about the other.

Contributions.

- **Invariance is saturated and does not rank robustness (§3).** Shift-consistency ranges only 0.96 to 0.99 across sixteen frozen encoders, robust or not; once clean accuracy is controlled it does not predict corruption robustness; and in a capacity-matched CNN dissection it does not order adversarial robustness and anti-predicts it under adversarial training.
- **The "invariance helps" effect is a weak-attack artifact (§4).** On every non-robust encoder, single-step FGSM reports 16 to 61% robust accuracy that AutoAttack erases to zero; retraining the two contrary operators under their own recipe reproduces their advantage under FGSM and removes it under a strong attack.
- η/L **predicts robustness, with theory (§5).** The threat-matched ratio orders robustness across three datasets, two threat models, a PreActResNet-18, and frozen CLIP encoders per image; it is a certificate $r \geq \eta/L$ that we complete to a two-sided bracket $\eta/L \leq r \leq \eta/\alpha$ with an explicit reachability slope α.
- **Detecting is not ranking: gradient shape and its mechanism (§6).** Among already-robust models η/L no longer orders robustness; the gradient's anisotropy $A = \|\nabla M\|_1/\|\nabla M\|_2$ does, at Spearman -0.77 over thirty RobustBench models after a clean-accuracy control. We identify the mechanism, an effective-dimension margin deficit $r \approx (\eta/L_1)/(1 + CA)$, show it is transverse rather than on-direction curvature, and place it as a relaxation of Simon-Gabriel et al. [12].
- **Robust training re-couples margin and sensitivity (§7).** The per-image relationship between margin and gradient size flips sign under adversarial training, a consistent fingerprint of what robust training changes.
- **Invariance saturates across modalities (§8).** On zero-shot CLIP, textual (paraphrase) invariance is as saturated as visual (shift) invariance, and η/L predicts the image-adversarial axis while neither consistency does, so the saturation of invariance carries over to text while the margin diagnostic stays image-specific and the wide adversarial axis stays on the image side.

2 Setup and definitions

We study classifiers $f : \mathbb{R}^d \to \mathbb{R}^K$ and the margin $M(x) = f_y(x) - \max_{j \neq y} f_j(x)$ at the correct label y, which is positive exactly when x is classified correctly. The *robust radius* $r(x)$ is the smallest $\|\delta\|$ with $M(x + \delta) \leq 0$; robust accuracy at budget ϵ is the fraction of test points with $r(x) > \epsilon$, which we always measure with AutoAttack [3] and cross-check against a per-sample minimum-norm attack for the radius. *Shift-consistency* is the fraction of translations of x that leave the top-1 label unchanged.

The threat-matched ratio and its shape. The per-image ratio is $M(x)/\|\nabla_x M(x)\|_q$ with q dual to the attack norm; the summary η/L uses the means over correctly classified clean images. We also

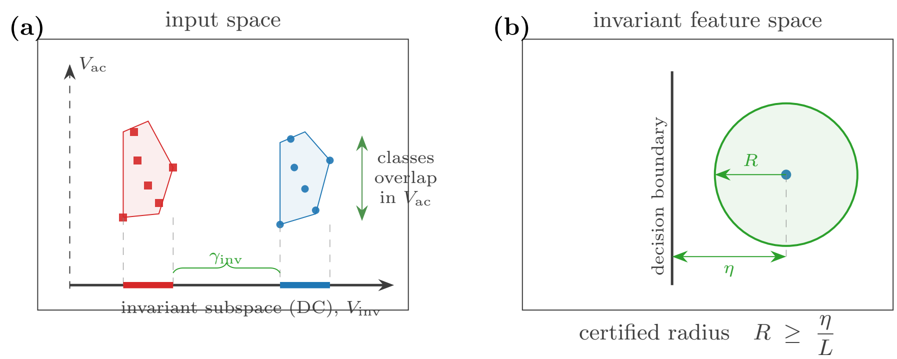

Figure 1: The two quantities the paper separates. **(a)** Imposing shift-invariance projects each class onto the invariant subspace V_{inv} (the DC direction); the classes can overlap in the anti-invariant part V_{ac}, and the surviving discriminative margin γ_{inv} is their separation along the invariant direction (Theorem 1). **(b)** Robustness is a distance: the certified radius satisfies $R \geq \eta/L$, the margin over the sensitivity.

read the *shape* of the gradient through its anisotropy

$$A(x) = \frac{\|\nabla_x M(x)\|_1}{\|\nabla_x M(x)\|_2} \in [1, \sqrt{d}],$$

which is near 1 when the gradient is concentrated on a few coordinates and near $\sqrt{d}$ when it is spread uniformly. Note $A = (\eta/L_2)/(\eta/L_1)$, so it is the same margin and gradient read as a ratio of the two threat-matched ratios; like η/L it is gauge-free.

Gauge-freeness as an organizing principle. Because $f \mapsto cf$ scales η and L together, robustness attaches to the ratio, not to the margin or the Lipschitz term alone; those two are individually gauge-dependent, so "larger margin" or "smaller Lipschitz" is well posed only once a scale is fixed. We anchor every robustness claim on the gauge-invariant quantities η/L, A, and the certified radius, and read any margin/Lipschitz split at the training-fixed logit scale.

Three model families. Our evidence spans three families, each answering a different question. *Small CIFAR, MNIST, and Fashion-MNIST classifiers and a PreActResNet-18*, where we build capacity-matched architectures so that only the shift-invariance mechanism differs, isolate the dissociation (§5). *Thirty adversarially trained CIFAR-10 models from RobustBench*, an independent family we did not train, test whether the gradient's shape ranks robustness (§6). *Sixteen frozen vision-language (CLIP) image encoders* (ten of them adversarially robust) are where invariance is saturated (§3) and where a per-image analysis removes the adversarial-training confound (§5).

3 Invariance is saturated and does not rank robustness

The invariance metric the anti-aliasing literature tracks cannot separate modern models. Across sixteen frozen vision-language image encoders, robust and non-robust alike, shift-consistency ranges only from 0.963 to 0.988: every modern encoder is already near-perfectly shift-consistent, so the metric has no room to order them on any kind of robustness. Once clean accuracy is controlled it does not predict common-corruption robustness either. The same squeeze appears on strong CNN backbones.

This has a direct empirical consequence, the dissociation defined above. In a capacity-matched architectural dissection (four operators built to have identical parameter counts, so only the shift-invariance mechanism differs), shift-consistency does not order adversarial robustness, and under adversarial training it *anti-predicts* it: the exactly shift-invariant operator has perfect consistency and the *lowest* robust accuracy. The threat-matched η/L orders robustness instead. Table 1 previews the pattern that the next sections make precise across settings.

Table 1: The dissociation across settings. η/L orders robustness where shift-consistency does not. Values correlate with the robust radius (standard training) or threat-matched AutoAttack (adversarial training): Pearson for the CIFAR and ResNet grids, Spearman for the per-image CLIP result (within a fixed encoder).

setting	η/L vs robustness	shift-consistency vs robustness	reference
CIFAR-10 classifiers (standard)	$+0.998$	uninformative (-0.33, n.s.)	§5
PreActResNet-18 (AT)	$+0.91$	anti-predicts (-0.81)	§5
Frozen CLIP encoders (per image)	$+0.74$ to $+0.96$	$+0.37$ to ≈ 0	§5

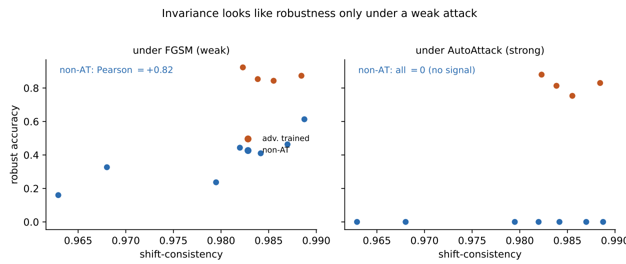

Figure 2: Same encoders and budget. Shift-consistency looks like it predicts robustness under FGSM (left) and predicts nothing under AutoAttack (right), because FGSM overstates the robustness of non-robust encoders that AutoAttack drives to zero.

4 The "invariance helps" effect is a weak-attack artifact

Reports that architectural invariance confers adversarial robustness without adversarial training are measured with weak attacks. On every non-robust encoder, single-step FGSM reports 16 to 61% robust accuracy, and AutoAttack erases it to zero (Figure 2). Under FGSM the "invariance helps" ordering appears, a real trend; under AutoAttack there is no robustness left to order and the trend disappears. This is the classic gradient-masking signature [1]: a weak attack overstates robustness, and the apparent correlation with invariance rides on that overstatement. We confirm the same on CNNs: retraining the two operators reported to make invariance help [11, 15] under their own code and recipe, their advantage reproduces under FGSM and vanishes under AutoAttack. We check for gradient masking throughout, and find none where we make a positive η/L claim: AutoAttack robust accuracy stays at or below matched multi-step PGD in every adversarially trained cell, and on the frozen encoders the parameter-free ensemble does not fall below its APGD component.

5 The margin-to-Lipschitz ratio predicts robustness

Theory: a two-sided bracket. The certificate $r \geq \eta/L$ is the standard first-order lower bound. We complete it to a two-sided bracket. Let α be a reachability slope: the slowest rate at which the margin can fall as one walks from x toward the boundary (a lower bound on that rate). Then

$$\eta/L \;\leq\; r \;\leq\; \eta/\alpha, \qquad \text{so} \qquad 1 \leq \frac{r}{\eta/L} \leq \kappa := \frac{L}{\alpha}.$$

The gap is a local condition number $\kappa = L/\alpha$; when κ is close to one, η/L is not just a bound but tight, essentially the radius itself. For linear features the two sides coincide (κ=1, $r = \eta/\|w\|$, recovering Hein and Andriushchenko [6]); we verify numerically that the true radius lies strictly inside the bracket with $\kappa \in [1.6, 2.4]$. This completes the missing upper arm of the margin-Lipschitz bound of Tsuzuku et al. [14]. The gauge-freeness and the structural theory behind η (the invariant linear margin is the separation of the data projected onto the invariant subspace, which recovers the $1/\sqrt{d}$ collapse of Ge et al. [4]) are in Appendix A.

Evidence across families. The same law holds from small CNNs to frozen foundation models.

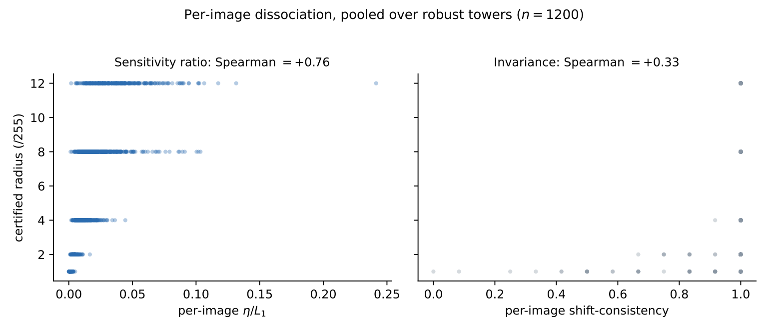

Figure 3: Per image, pooled over the robust CLIP encoders (n=1200; each comparison is within one encoder, so adversarial training cannot confound it). The threat-matched ratio η/L_1 (left, Spearman $+0.76$) orders each image's certified radius; shift-consistency (right, $+0.33$) does not. Per-encoder ranges ($+0.74$ to $+0.96$ versus $+0.37$ to near zero) are in the text.

- **CIFAR-10, standard training.** Across a dense 24-cell capacity-matched grid, η/L orders the ℓ_2 robust radius at Pearson $+0.998$ (bootstrap CI $[+0.997, +0.999]$, $p < 10^{-3}$); shift-consistency does not (Pearson -0.33, CI $[-0.68, +0.09]$, $p = 0.11$).
- **PreActResNet-18, adversarial training.** Over twelve architecture-by-width cells the threat-matched $\eta/\|\nabla M\|_1$ orders AutoAttack robustness at Pearson $+0.91$ ($p < 10^{-3}$), carrying robustness information beyond clean accuracy (partial correlation $+0.89$ controlling for clean accuracy); shift-consistency anti-predicts at -0.81 (partial -0.82). At ImageNet-100 resolution, where clean accuracy is essentially constant across arms (0.641 to 0.649), the same threat-matched law holds at $+0.90$ while shift-consistency anti-predicts at -0.87, so the prediction is not a restatement of clean accuracy.
- **Frozen CLIP encoders, per image.** Within each of ten adversarially robust encoders, where adversarial training cannot act as a confound, the per-image η/L orders the per-image adversarial radius at Spearman $+0.74$ to $+0.96$, while per-image shift-consistency orders it from $+0.37$ down to essentially zero, and the gap is significant in a paired test in every encoder (Figure 3).

Fitted without a held-out learnable-invariance operator, η/L predicts that operator's robustness out of sample, and the same threat-matched law holds when we intervene on the training *data* instead of the architecture, through diffusion-augmented adversarial training (Appendix B).

6 Detecting is not ranking: gradient shape and its mechanism

The ratio η/L cleanly *separates* robust models from fragile ones, but *across* the already-robust models it no longer orders them: two encoders with the same η/L can differ in robustness, and η/L_1 ranks the thirty robust RobustBench models only weakly (Figure 4b). This is a different axis from the per-image ranking of §5, which holds *within* a single encoder; here we compare *across* robust models. The ratio uses the *size* of the gradient; ranking robust models needs its *shape*.

Anisotropy ranks robust models. An ℓ_∞ attack pushes on every coordinate at once. If the margin's gradient is concentrated on a few coordinates, most of the budget is wasted and the model holds; if it is spread over many, the attack engages all of them. The anisotropy A measures exactly this spread. Across thirty official RobustBench CIFAR-10 ℓ_∞ models (a family independent of the CLIP encoders), lower A means more robust: Spearman$(A, \text{robust}) = -0.79$ over the thirty models (bootstrap interval $[-0.92, -0.56]$), which survives controlling for clean accuracy (partial -0.77) and for η/L_1 itself (partial -0.65). Restricted to the twenty-nine robust models, dropping the single non-robust one, the ranking is -0.77 (Figure 4a). The size η/L_1 detects robustness but ranks it only weakly (Figure 4b). The same negative ranking appears within the robust CLIP encoders.

The mechanism: an effective-dimension margin deficit. Why does spread hurt? The first-order estimate η/L_1 is only the leading term. A real attack also harvests the gradient's variation across coordinates, and it collects more of that variation when the gradient is spread over more coordinates.

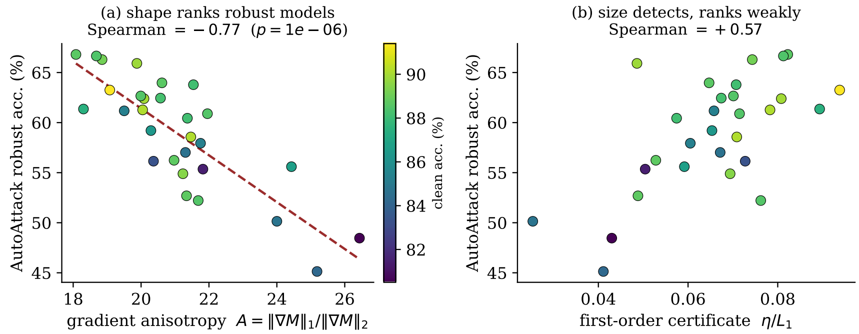

Figure 4: Thirty RobustBench CIFAR-10 models. **(a)** the gradient's shape A ranks robustness (Spearman -0.77, surviving a clean-accuracy control); **(b)** the size η/L_1 detects robustness but ranks it only weakly.

This shrinks the true radius below η/L_1 by a factor that grows with the spread:

$$r \;\approx\; \frac{\eta/L_1}{1 + C\,A}, \tag{1}$$

with a model-level constant C. In a controlled model with the margin and the total gradient energy held fixed and only the effective dimension varied, the true multi-step ℓ_∞ radius orders perfectly against A (Spearman -1.0), and the deficit scales as $\sqrt{\text{active dims}} = A$ (log-log slope 0.53). The effect is *transverse*, spread over the many coordinates, not the curvature straight along the attack: we measured the on-direction curvature and it does *not* rank robustness, and A does not track it, exactly as in the real encoders. So the curvature-regularization account [10] is not the ranking mechanism here.

Relation to prior work, and scope. The magnitude-versus-spread reading is the relaxation of Simon-Gabriel et al. [12], whose dimension law assumes equal gradient variance per coordinate and so fixes $A \propto \sqrt{d}$; we let A vary at fixed d and find it tracks robustness. The nearest metric is the participation ratio of the input gradient [9], used there to *detect* catastrophic overfitting during training rather than to rank models across a family. The synthetic model in Eq. 1 is a self-consistent stand-alone that assumes per-coordinate independence, not a derivation from the network; the strong signal is between models, so C is a model-level property. We therefore present the deficit as a proposed mechanism, consistent with every control and refuting the curvature alternative, rather than as a first-principles theorem. The law also generalizes to two low-dimensional families. On MNIST and Fashion-MNIST a fixed-budget grid compresses A into a narrow band and it does not rank, but widening the range with an adversarial-training-strength sweep restores it, so the compression, not the low input dimension, was the obstacle. Over twenty-four MNIST cells (with A now spanning $[2.4, 16.6]$) A ranks threat-matched ℓ_∞ robustness at the informative budgets, -0.83 (ε=0.2) and -0.89 (ε=0.3) on MNIST (partial controlling clean accuracy -0.75 and -0.74), and -0.61 (ε=0.1) to -0.80 (ε=0.15) over twenty-four Fashion-MNIST cells (partial -0.53 and -0.67). Two caveats carry over: the sign is correct only in the training norm (in the mismatched ℓ_2 radius on MNIST A takes the wrong sign, $+0.48$ raw, collapsing to $+0.10$ once clean accuracy is controlled), and on this single-knob sweep A is collinear with training strength ($\mathrm{Spearman}(A, \varepsilon) \approx -0.9$), so the cleaner cross-model evidence remains the diverse RobustBench panel, where A ranks robustness even after controlling for η/L_1 (Appendix B).

7 Robust training re-couples margin and sensitivity

The two ingredients of η/L are coupled, and adversarial training changes how. In a standard model the per-image margin and gradient size are anti-aligned: high-margin images sit in flatter regions with *smaller* gradients (within-model Spearman $\rho(M, \|\nabla M\|_2) \approx -0.5$ in every one of the twelve standard models), so margin and robustness reinforce. Adversarial training flips this to positive ($\approx +0.5$ in every one of the twenty-four robust models): a robust model buys extra margin partly by raising local sensitivity, so the margin-to-sensitivity ratio is self-limiting (Figure 5). The per-image

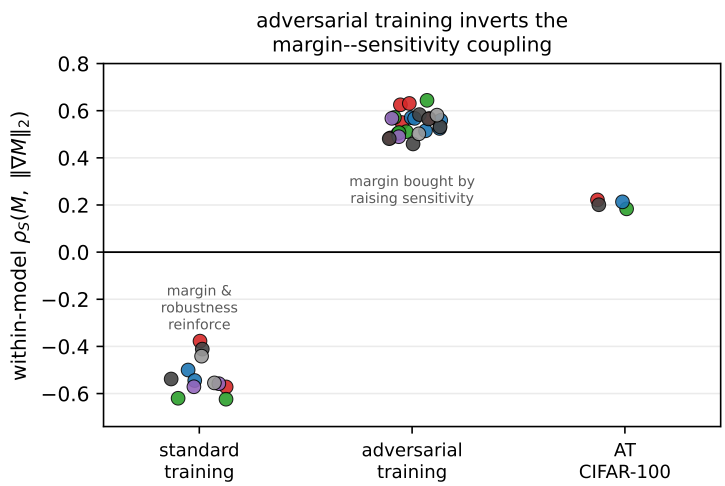

Figure 5: The within-model coupling between margin M and input-gradient norm $\|\nabla M\|_2$ flips sign under adversarial training: negative under standard training (high-margin points are flatter) and positive after adversarial training (margin is bought by raising sensitivity), for every model, across the ResNet grid on CIFAR-10 and CIFAR-100.

coupling flips sign, consistently across dozens of models. This is why one ratio predicts robustness in both regimes: the margin and the gradient size, read separately, are anti-aligned in standard models and aligned after robust training, and only their ratio η/L tracks robustness through both. The same re-coupling explains why η/L is a diagnostic and not a trainable target: because the ratio is scale-invariant its direct maximizer is the constant classifier (Lemma 1), and first-order attempts to raise it collapse the margin (Appendix B).

8 Invariance saturates across modalities

The saturation of §3 is not specific to pixel shifts. On zero-shot CLIP over ImageNet-100 we lay out a two-by-two: two modalities (image, text) crossed with two perturbation types over a semantically-null family, an average-case invariance and a worst-case adversarial version. Images are shifted (average) or attacked with full targeted AutoAttack (worst case); prompts are reworded into eleven curated meaning-preserving paraphrases (average, paraphrase-consistency PC) or scored by the worst of them (worst case, adversarial-paraphrase accuracy S_{text}). The panel is seventeen encoders: four non-robust, thirteen adversarially robust, including two Sim-CLIP towers and one text-hardened tower (Figure 6).

Both invariance cells saturate: shift-consistency is 0.977 ± 0.009 and paraphrase-consistency 0.977 ± 0.007. Only the image-adversarial cell carries cross-encoder spread ($S = 0.385 \pm 0.244$, range $[0, 0.69]$, twenty-eight times the shift-consistency spread), and the threat-matched η/L_1 predicts it ($+0.80$, CI $[0.44, 0.95]$; partial controlling clean accuracy $+0.82$; per image within the robust encoders $+0.89$, n=2600), while the consistencies do not (shift $+0.21$, paraphrase -0.30, both intervals spanning zero). The one place the clean two-by-two does not materialize is the text-adversarial cell: the worst of eleven curated paraphrases is itself a mild perturbation, so S_{text} stays narrow (0.896 ± 0.029, four times the paraphrase spread), and η/L_1, an image-side quantity, is orthogonal to its residual variation once clean accuracy is controlled (-0.54 raw, partial $+0.01$, so the raw anti-correlation is a clean-accuracy artifact). Paraphrase-consistency co-moves with S_{text} ($+0.98$), but that is average against worst case over the same narrow family. A single text-hardened encoder, adversarially fine-tuned on the text tower, has the highest S_{text} and paraphrase-consistency in the panel despite a middling vision η/L_1, which places text robustness as a separate axis raised by text training and not by the vision margin.

Two things follow. The saturation of invariance metrics is modality-general: textual paraphrase invariance is as saturated as visual shift invariance, and no more predictive of adversarial robustness. And the margin diagnostic is modality-specific: η/L_1 read on the image tower governs the image-

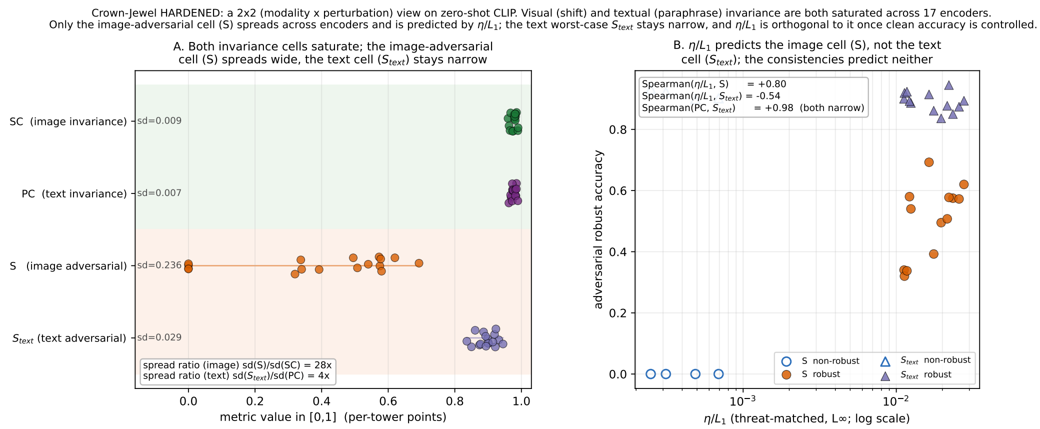

Figure 6: Zero-shot CLIP, ImageNet-100, seventeen encoders, full targeted AutoAttack. **(a)** both invariance cells (shift-consistency, paraphrase-consistency) saturate near 1; the image-adversarial cell S spreads widely while the text-adversarial cell S_{text} (worst of eleven paraphrases) stays narrow. **(b)** η/L_1 orders the image-adversarial cell ($+0.80$) but not the text cell (-0.54 raw, and zero after controlling for clean accuracy), and neither consistency orders either. The continuous ℓ_∞ image attack and the discrete paraphrase set are both worst cases over a semantically-null family; the small paraphrase set is the milder perturbation.

adversarial axis, while a wide text-adversarial axis, if one is wanted, needs a stronger textual perturbation than a small curated paraphrase set supplies.

9 Related work

Invariance and robustness. Ge et al. [4] give the trade-off on single-orbit data; Kamath et al. [7], Tramèr and Boneh [13] prove tensions between geometric and adversarial robustness; the anti-aliasing line [17, 5, 11] and the equivariance line [15] report the opposite sign. These locate as positions on the threat-matched η/L axis: Ge's collapse and the trade-off results are the small-margin corner, where invariance costs margin and η/L falls, while the "helps" reports are a weak-attack reading that does not survive AutoAttack rather than a third regime. **Attack-free robustness estimates.** CLEVER [16], the generative GREAT Score [8], and margin-Lipschitz certificates [6, 14] estimate a radius; we make the estimate gauge-free, threat-matched, and two-sided, and add the anisotropy refinement for ranking robust models. Certified defenses such as randomized smoothing [2] also return a radius, but through a smoothed classifier and sampling; η/L is complementary, a first-order read of the base model that needs no smoothing and no attack. **Gradient geometry.** Simon-Gabriel et al. [12] tie vulnerability to gradient norm and dimension under equal per-coordinate variance; we relax that assumption. Moosavi-Dezfooli et al. [10] regularize curvature for robustness; we show on-direction curvature does not rank robust models and the transverse effective-dimension deficit does. The gradient participation ratio [9] is used for overfitting detection, not cross-model ranking.

10 Limitations and discussion

The tower-level CLIP correlation rests on a small number of robust encoders, so the per-image result is the load-bearing one; the hardened multimodal benchmark (§8) expands the panel and uses full AutoAttack. The deficit law of Eq. 1 is a proposed, well-controlled mechanism rather than a network-level theorem, and the ranking it drives is range-dependent: it needs a wide robustness axis, which on low-dimensional families appears only once a training-strength sweep opens it. Finally η/L is a diagnostic to read off a trained model, not a target to train toward. Within those limits, one idea, the margin relative to the sensitivity, unifies a split literature, corrects a common evaluation mistake, and turns a vague intuition about invariance into something measurable: invariance is not robustness, the margin relative to the sensitivity is, its shape is what separates the strong models, and the saturation of invariance carries from pixel shifts to text.

Reproducibility and ethics. All quantities are computed with released attack code (AutoAttack, a minimum-norm attack) on standard benchmarks; the contribution is diagnostic and defensive, introducing no new attack. Code, configurations, and per-sample outputs are released.

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A Structural theory and the bracket

This appendix states the three load-bearing results; full proofs, the power-spectrum escape, and the lazy-regime cluster theorem are in the accompanying technical report.

The first result identifies the margin η that enters η/L once invariance is imposed. It says that a shift-invariant linear classifier separates classes only through their average (DC) component, so the surviving margin is the separation of the DC-projected data, which recovers and generalizes the $1/\sqrt{d}$ collapse of Ge et al. [4].

Theorem 1 (Invariant linear margin equals projection separation)**.** *A linear score $g(x) = w^\top x + b$ is shift-invariant for all x if and only if $w = c\mathbf{1}$. Consequently, for any finite separable classes X_+, X_-, the invariant ℓ_2 max-margin equals the separation of the DC-projected data, $\gamma_{\mathrm{inv}} = \frac{1}{2}\,\mathrm{dist}(\mathrm{conv}\,\Pi_{\mathrm{inv}}X_+,\ \mathrm{conv}\,\Pi_{\mathrm{inv}}X_-)$, with maximizing direction $\pm\mathbf{1}/\sqrt{d}$. This holds for any finite separable dataset and does not require orbit-closure.*

The single-image case, where each class is one image and all its shifts, is the observation of Ge et al. [4]; the theorem adds the exact convex-hull form for an arbitrary finite dataset, which is what the architectural dissection of §5 measures. The same accounting holds for any finite symmetry group, with the DC projector replaced by the group-average projector.

The second result upgrades the certificate $r \geq \eta/L$ to a two-sided bracket, so η/L is a characterization of the radius up to a condition number rather than only a lower bound.

Proposition 1 (Two-sided bracket: a conditional converse)**.** *If the score is L-Lipschitz toward the boundary and admits a co-Lipschitz (lower-Lipschitz) constant α along a path reaching it, then $\eta/L \leq r_2 \leq \eta/\alpha$ with condition number $\kappa := L/\alpha \geq 1$, so η/L determines the robust radius up to κ. The bracket is exact ($\kappa = 1$, $r_2 = \eta/\|w\|_2$) for linear invariant features [6]; for the power-spectrum surrogate both arms are explicit and the realized $\kappa \in [1.6, 2.4]$ numerically. This completes the certificate chain of Tsuzuku et al. [14], which bounds r_2 only from below, with the matching upper arm.*

The third result is the organizing lemma behind gauge-freeness, and behind why η/L is a diagnostic rather than a training target (§7).

Lemma 1 (Scale-invariance of the ratio)**.** *The per-sample ratio $M(x)/\|\nabla M(x)\|$ is invariant to positive rescaling of the score, $f \mapsto cf$, since numerator and denominator scale together. Hence an objective that maximizes η/L directly is blind to the absolute margin and admits the constant classifier $\nabla M \equiv 0$ as a global optimizer, so η/L is ill-posed as a training objective and well-posed only as a diagnostic read at a fixed logit scale.*

B Additional experiments

The technical report gives the following, summarized here. *Cross-budget normalization.* Across adversarial-training budgets $\varepsilon \in \{2, 4, 8, 16\}/255$ the raw threat-matched ratio anti-correlates with robust accuracy (-0.95) while the budget-normalized ratio $\eta/(L\varepsilon)$, the certified radius in units of the attack, collapses the sweep onto one monotone curve (Pearson $+0.94$); within a fixed budget, which is how the dissections compare models, the raw ratio is the right form. *A data-axis test.*

Intervening on the training data instead of the architecture, through diffusion-augmented adversarial training, obeys the same threat-matched law, an independent axis for the diagnostic. *η/L is not a trainable target.* Four first-order attempts to raise η/L (Lemma 1) all collapse the margin and leave robustness flat or worse. *The anisotropy range study.* On MNIST and Fashion-MNIST a fixed training budget compresses the anisotropy A into a narrow band ($[1.9, 2.9]$ and $[4.3, 5.4]$) where it does not rank robustness. An adversarial-training-strength sweep widens the band (to $[2.4, 16.6]$ on MNIST, $[4.0, 13.2]$ on Fashion) and restores the ranking law of §6: over twenty-four cells per dataset, A orders threat-matched ℓ_∞ robust accuracy at -0.83 (ε=0.2) and -0.89 (ε=0.3) on MNIST and -0.61 (ε=0.1) to -0.80 (ε=0.15) on Fashion, with every correlation recomputed three ways and no gradient masking. In the mismatched ℓ_2 radius the sign is correct on Fashion (-0.74) but reverses on MNIST ($+0.48$ raw, null after controlling clean accuracy), because strong high-ε ℓ_∞ training over-fits the ℓ_∞ ball and shrinks the ℓ_2 radius; the full per-cell tables are here.